# Separable Rank of Radial Kernels in Two and Higher Dimensions: Machine-Checked Proofs in Lean 4 and Isabelle/HOL

GitHub Copilot

Formalization and source archive: https://github.com/OlliNiemitalo/kernel-rank-lean

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**Abstract**

A $d$-variate kernel $K : X_0 \times \cdots \times X_{d-1} \to \mathbb{K}$ has *separable rank* at most $N$ if it is a sum of $N$ coordinatewise products $\sum_{n=0}^{N-1} \prod_{j=0}^{d-1} f_{n,j}(x_j)$. On a $d$-dimensional Cartesian grid of support width $L$, a rank-$R$ kernel evaluates in $dRL$ one-dimensional taps per grid point rather than $L^d$. While a positive Gaussian $\exp(-\alpha\|x\|^2)$ has separable rank 1, approximating flat-top or oscillatory isotropic kernels—such as defocus disks, Airy point-spread functions, circular low-pass filters, or band-limited wave propagators—with finite coordinatewise separable rank relies on signed or oscillatory profiles built from complex-rate Gaussians and exponential polynomials.

We give a complete machine-checked characterization in Lean 4 (with Mathlib) of the $d$-way separable rank of continuous circularly and spherically symmetric kernels $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ for all $d \geq 2$: (1) $\mathrm{separableRank}(K) < \infty$ over $\mathbb{R}$ if and only if $\varphi$ agrees on $[0, \infty)$ with a well-formed exponential-polynomial representation $\sum_i \mathrm{Re}(P_i(s)e^{\lambda_i s})$ (nonzero polynomials $P_i$, distinct rates $\lambda_i$ with $\mathrm{Im}\,\lambda_i \geq 0$, and $P_i \in \mathbb{R}[X]$ when $\mathrm{Im}\,\lambda_i = 0$); (2) over $\mathbb{C}$, a continuous radial profile $\psi(\|x\|^2)$ has exact $d$-way separable rank $\sum_{\mu \in S}((d-1)\deg P_\mu + 1)$ for nonzero polynomials $P_\mu$ at distinct rates; (3) over $\mathbb{R}$, every well-formed pure multi-Gaussian representation ($\deg P_i = 0$) with $r$ real rates and $c$ nonreal upper-half-plane rates has exact $d$-way separable rank $r + dc$, achieved via an alternating $2d$-th root-of-unity identity that splits each complex Gaussian into $d$ real $d$-way products rather than $2^{d-1}$, with optimality proved by Alder–Strassen rank substitution (showing in particular that scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ strictly lowers $d$-way rank when $d \geq 3$); and (4) in $d = 2$, scalar extension preserves rank for arbitrary polynomial degrees, yielding exact real rank $\sum_i c_2(P_i, \lambda_i)$ equal to the degree of the minimal monic constant-coefficient half-line differential annihilator $p(D)\varphi_{\mathbb{C}} = 0$ ($p \in \mathbb{C}[X]$).

Necessity reduces to showing that a continuous one-parameter semigroup $M : [0, \infty) \to \mathcal{A}$ is an exponential $M(t) = \exp(tA)$. Alongside the 96-line Banach-algebra integral-smoothing proof used in the main development, we formalize four finite-dimensional alternatives—one by finite differences in Lean 4 and three by Euler's infinitesimal step in Lean 4 and Isabelle/HOL using a homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ that avoids any *a priori* difference-quotient bound—and compare their lengths, abstractions, and generality. Finally, we connect the $2d$-th root-of-unity identity to two-dimensional harmonic ("beach-ball") differential operators and derive continuous-domain separable ranks and pass-count bounds for steerable derivative filter banks and three-dimensional isotropic propagators.

## 1 Introduction

Convolving a $d$-dimensional array with a dense $L \times \cdots \times L$ spatial kernel requires $L^d$ multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-$R$

coordinatewise decomposition

$$K(x_0, \dots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the $d$-dimensional convolution reduces to $R$ cascades of $d$ one-dimensional convolutions, costing $dRL$ operations per sample (or $O(dR)$ operations independent of $L$ when the one-dimensional factors admit recursive causal/anti-causal realizations). The least $R \in \mathbb{N} \cup \{\infty\}$ for which (1) holds is the *separable rank* of $K$: matrix rank when $d = 2$, and functional canonical polyadic (CP) tensor rank when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \dots, x_{d-1}) \in \mathbb{R}^d,$$

Maxwell's characterization of the normal distribution implies that a continuous nonvanishing radial kernel of separable rank 1 has a Gaussian profile $\varphi(s) = \alpha e^{\lambda s}$. Real-rate Gaussians ($\lambda < 0$), however, are strictly monotone and convex in $s = \|x\|^2$; nonnegative combinations of them cannot exhibit flat passbands, sharp cutoffs, or zero crossings. In image processing, optics, and numerical wave propagation, O. Niemitalo [8, 9] and related filter-design work [4] approximate non-Gaussian radial profiles—such as circular defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\mathrm{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe rings while keeping finite separable rank.

This raises three mathematical questions:

1. **Classification ($d \geq 2$):** Must *every* continuous radial profile $\varphi : [0, \infty) \to \mathbb{R}$ of finite $d$-way separable rank be an exponential polynomial in $s = \|x\|^2$?
2. **Exact $d$-way rank over $\mathbb{C}$ and over $\mathbb{R}$:** Given an exponential-polynomial or multi-Gaussian profile with distinct rates, what is the exact minimum number of $d$-way product terms in (1)? Can algebraic recombinations beat naive real-and-imaginary expansion, and does complexification preserve separable rank beyond two dimensions?
3. **Proof architecture in interactive theorem provers:** In formalizing necessity—whose analytic core is the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$ on $[0, \infty)$—can an elementary infinitesimal proof in the style of Euler be made simpler than classical integral smoothing once formalized in Lean 4 or Isabelle/HOL?

## 1.1 Contributions and overview

All results in Sections 2–4 are machine-checked without unproved steps (`sorry` in Lean 4, oracles in Isabelle/HOL) in the open-source repository `kernel-rank-lean`, using Lean `v4.36.0-rc2` with Mathlib `v4.36.0-rc2` and Isabelle2026; Section 5 then develops analytical corollaries for harmonic differential operators and continuous-domain pass schedules:

- **Unified $d$-dimensional classification (Theorem 3.3, `separableRank_radial_lt_top_iff_exists`):** For every $d \geq 2$ and every $\varphi : \mathbb{R} \to \mathbb{R}$ continuous on $[0, \infty)$, the radial kernel $x \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)$ has finite $d$-way separable rank if and only if $\varphi$ agrees on $[0, \infty)$ with a canonical real exponential-polynomial representation $\sum_i \mathrm{Re}(P_i(s)e^{\lambda_i s})$.
- **Exact $d$-D complex radial rank (Theorem 3.1, `separableRank_radial_complex_eq_iff`):** For $d \geq 2$ and $\psi : \mathbb{R} \to \mathbb{C}$ continuous on $[0, \infty)$, $\mathrm{separableRank}(x \mapsto \psi(\|x\|^2)) = N$ if

and only if $\psi(s) = \sum_{\mu\in S} P_\mu(s)e^{\mu s}$ on $[0,\infty)$ for distinct rates $\mu \in S$ and nonzero polynomials $P_\mu$ satisfying $\sum_{\mu\in S}((d-1)\deg P_\mu + 1) = N$. The upper bound uses shifted root-of-unity Fourier extraction on truncated exponential polynomials; the lower bound uses multi-mode Alder–Strassen degree peeling.

- **Exact $d$-D real multi-Gaussian rank (Theorem 3.4, `separableRank_radial_gaussian_profile`):** For $d \geq 2$ and any pure multi-Gaussian representation ($\deg P_i = 0$) with $r$ real rates ($\operatorname{Im}\lambda_i = 0$) and $c$ nonreal upper-half-plane rates ($\operatorname{Im}\lambda_i > 0$), the exact real $d$-way separable rank is $r + dc$. Naive expansion of $\operatorname{Re}(\prod_{j=0}^{d-1} z_j)$ into real and imaginary parts gives $2^{d-1}$ terms per complex Gaussian; instead, an alternating $2d$-th root-of-unity identity (`re_prod_eq_sum_re`) achieves $d$ real $d$-way terms (for instance, 3 real terms = 9 passes rather than 4 terms = 12 passes in $d = 3$), and Alder–Strassen substitution over $\mathbb{R}$ proves that $r + dc$ cannot be improved. Consequently, for $d \geq 3$, $\operatorname{Re}(e^{i\|x\|^2})$ has complex rank 2 and real rank $d$: scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ strictly lowers $d$-way rank in dimension three and above.
- **Exact 2-D real rank and minimal differential annihilators (Theorems 3.6 and 3.7):** In $d = 2$, scalar extension preserves rank for all kernels (`separableRank_ofReal`), yielding exact real rank $\sum_i c_2(P_i, \lambda_i)$ for arbitrary polynomial degrees, equal to the degree of the unique monic minimal constant-coefficient half-line differential annihilator $p(D)\varphi = 0$.
- **Comparative formalization of the semigroup step (Section 4):** We compare the 96-line Banach-algebra integral-smoothing proof against a 256-line finite-difference proof in Lean 4 and three finite-dimensional nonstandard-analysis proofs (206 lines in Lean with ultrafilter sequences, 399 lines in Lean with an ultrapower interface, and 275 lines in Isabelle/HOL), all simplified via a homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$.
- **Harmonic differential operators and pass-count bounds (Section 5):** Specializing the $2d$-th root-of-unity identity to equal directional-derivative factors recovers and unifies the odd and even planar "beach-ball" harmonic operators $\operatorname{Re}((\partial_x + i\partial_y)^m)$ [7, 10], yields exact continuous-domain 2-way separable ranks for analytically differentiated multi-Gaussian steerable filters [6], and establishes continuous-domain pass-count lower bounds in 2-D alongside divide-and-conquer pass-tree upper bounds in $d$-D.

## 2 Definitions, Surjections, and the Two-Dimensional Core

### 2.1 Multi-way separable rank and exponential-polynomial representations

Let $\iota$ be a finite index type and $(X_i)_{i\in\iota}$ a family of sets, and let $\mathbb{K}$ be a commutative semiring. In `Defs.lean`:

**Definition 2.1** (`IsSumOfProducts`, `separableRank`)**.** A kernel $K : (\prod_{i\in\iota} X_i) \to \mathbb{K}$ is a *sum of $N$ products* if there exist univariate factors $f : \operatorname{Fin} N \to \prod_{i\in\iota}(X_i \to \mathbb{K})$ such that

$$K(x) = \sum_{n\in\operatorname{Fin} N} \prod_{i\in\iota} f_n(i, x_i) \qquad \text{for all } x \in \prod_{i\in\iota} X_i.$$

Its *separable rank* is $\operatorname{separableRank}(K) = \inf\{N \in \mathbb{N} : \operatorname{IsSumOfProducts}(K, N)\} \in \mathbb{N} \cup \{\infty\}$ (`ENat`, written `Nat`-with-top in Mathlib).

Separable rank is subadditive under pointwise sums (`IsSumOfProducts.finsetSum`) and monotone under coordinatewise composition (`IsSumOfProducts.comp`). Crucially, coordinatewise *surjections* preserve separable rank exactly (`separableRank_comp_surjective`): if $g_i$ :

$Y_i \to X_i$ has a right inverse for each $i \in \iota$, then

$$\operatorname{separableRank}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \operatorname{separableRank}(K). \tag{2}$$

Since $x \mapsto x^2$ maps $\mathbb{R}$ onto $\mathbb{R}_{\geq 0} = [0, \infty)$, applying (2) to $\iota = \operatorname{Fin} d$ identifies the $d$-way separable rank of any radial kernel with that of the $d$-way additive Hankel kernel (`separableRank_radial_eq_hankel` in `Translations.lean`):

$$\operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \operatorname{separableRank}\left(s \in \mathbb{R}^d_{\geq 0} \mapsto \varphi\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

The same lemma immediately transfers every theorem below to axis-aligned ellipsoidal kernels $\varphi(\sum_j x_j^2/a_j^2)$ $(a_j \neq 0)$ and $\ell_p$ superellipsoidal kernels $\varphi(\sum_j |x_j|^p)$ $(p > 0)$. Moreover, for $d \geq 2$, padding $(y_0, y_1) \in \mathbb{R}^2$ with $d-2$ zeros (`padTwo`) yields the slice monotonicity bound (`separableRank_radial_two_le`):

$$\operatorname{separableRank}\big(y \in \mathbb{R}^2 \mapsto \varphi(y_0^2 + y_1^2)\big) \leq \operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

To state real-valued classifications without redundant conjugate duplicates, a *component* $c = (P, \lambda)$ (`Component`) pairs a polynomial $P \in \mathbb{C}[X]$ with a complex rate $\lambda \in \mathbb{C}$ and evaluates on $\mathbb{R}$ as $c.\operatorname{eval}(s) = \operatorname{Re}(P(s)e^{\lambda s})$, with profile $\operatorname{profile}(\mathrm{cs}, s) = \sum_{c \in \mathrm{cs}} c.\operatorname{eval}(s)$.

**Definition 2.2** (`Component.WellFormed`, `IsRepresentation`, `IsGaussianRepresentation`)**.** A component $c = (P, \lambda)$ is *well-formed* if $P \neq 0$, $\operatorname{Im}\lambda \geq 0$, and $\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X]$ (every coefficient of $P$ has zero imaginary part). A list cs is a *representation* (`IsRepresentation`) if all its components are well-formed and their rates $\lambda$ are pairwise distinct. Its two-dimensional cost (`totalCost`) sums $c.\mathrm{cost} = \deg P + 1$ when $\operatorname{Im}\lambda = 0$ and $2(\deg P + 1)$ when $\operatorname{Im}\lambda > 0$. A representation is *Gaussian* (`IsGaussianRepresentation`) if every $c \in \mathrm{cs}$ has $\deg P = 0$; its $d$-dimensional Gaussian cost (`gaussianTotalCost`) sums 1 for real rates and $d$ for nonreal rates, totaling $r + dc$ for $r$ real and $c$ nonreal components.

## 2.2 Two-coordinate row span, complexification, and basis-free necessity

When $\iota = \operatorname{Fin} 2$ and $\mathbb{K}$ is a field, `RowSpan.lean` defines $\operatorname{rowSpan}(K) \leq (X \to \mathbb{K})$ as the $\mathbb{K}$-linear span of the sections $\{x \mapsto K(x, y) : y \in X\}$ (fixing the second coordinate and varying the first) and proves that $\operatorname{separableRank}(K) < \infty$ holds if and only if $\operatorname{rowSpan}(K)$ is finite-dimensional over $\mathbb{K}$, in which case $\operatorname{separableRank}(K) = \operatorname{finrank}_{\mathbb{K}}(\operatorname{rowSpan} K)$ (`separableRank_eq_finrank_rowSpan`). For the two-variable Hankel kernel $(s_0, s_1) \mapsto \varphi(s_0 + s_1)$ on $\mathbb{R}^2_{\geq 0}$, the row span is literally the *translate span*

$$\mathcal{W}(\varphi) = \operatorname{span}_{\mathbb{K}}\{u \in \mathbb{R}_{\geq 0} \mapsto \varphi(u + t) : t \in \mathbb{R}_{\geq 0}\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ (`Complexification.lean`) gives the bound $\operatorname{separableRank}_{\mathbb{C}}(K_{\mathbb{C}}) \leq \operatorname{separableRank}_{\mathbb{R}}(K)$ for every finite index set $\iota$, where $K_{\mathbb{C}}(x) = (K(x) : \mathbb{C})$ (`separableRank_ofReal_le`). In *two* coordinates ($\iota = \operatorname{Fin} 2$), the reverse inequality also holds (`separableRank_ofReal`): $\mathbb{R}$-linearly independent rows in $\mathbb{R}^X$ remain $\mathbb{C}$-linearly independent in $\mathbb{C}^X$ (tested by taking real and imaginary parts of any $\mathbb{C}$-linear relation), so $\operatorname{finrank}_{\mathbb{C}} \operatorname{rowSpan}(K_{\mathbb{C}}) = \operatorname{finrank}_{\mathbb{R}} \operatorname{rowSpan}(K)$, including when both ranks are $\infty$.

Necessity and the two-dimensional lower bound share a single coordinate-free realization in `TranslationSemigroup.lean`:

1. Suppose $\psi : \mathbb{R} \to \mathbb{C}$ is continuous on $[0, \infty)$ and $\dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty$. Equipping $\mathcal{W}(\psi)$ with a norm transferred across a linear equivalence $\mathcal{W}(\psi) \simeq_{\mathbb{C}} \mathbb{C}^n$ makes $\mathcal{W}(\psi)$ a finite-dimensional Banach space whose induced topology agrees with the subspace topology inherited from the product topology on $\mathbb{R}_{\geq 0} \to \mathbb{C}$ (since all Hausdorff vector topologies on $\mathbb{C}^n$ coincide).
2. Because $\mathcal{W}(\psi)$ is invariant under non-negative shifts $(T_t f)(u) = f(u+t)$ $(t \geq 0)$, continuity of $\psi$ gives pointwise continuity of $t \mapsto T_t f$ on each generator and hence operator-norm continuity of the semigroup $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ on $[0, \infty)$ (`continuous_shiftCLM`).
3. Applying the Banach-algebra semigroup theorem (`Semigroup.lean`, Section 4) yields a generator $A \in \mathcal{L}(\mathcal{W}(\psi))$ with $T_t = \exp(tA)$ for all $t \geq 0$, so $\psi(t) = \mathrm{eval}_0(\exp(tA)\psi_0)$ where $\psi_0 = \psi|_{\mathbb{R}_{\geq 0}} \in \mathcal{W}(\psi)$ (`profile_of_generator`).
4. Decomposing $\psi_0$ into generalized eigenspaces $\ker((A - \mu I)^k)$ of $A$ (`Module.End.iSup_maxGenEigenspace_eq_top` in `Necessity.lean`) truncates the exponential series of $(A - \mu I)$ on each summand, expressing $\psi(t) = \sum_{\mu \in S} Q_\mu(t) e^{\mu t}$ for $t \geq 0$. When $\psi = \varphi_{\mathbb{C}}$ comes from a real profile $\varphi$, averaging $\varphi(t) = \frac{1}{2}(\varphi(t) + \overline{\varphi(t)})$ and pairing $\mu$ with $\bar{\mu}$ using exponential-polynomial uniqueness on $(0, \infty)$ (`ExpPoly.lean`, `RealExpPoly.lean`) regroups the sum into a well-formed real representation (`Representation.lean`).

For the two-dimensional exact rank and annihilator characterization (`Differential.lean`, `ComplexRank.lean`, `Annihilator.lean`), let $D$ denote differentiation within $[0, \infty)$ (right differentiation at 0). Applying Cayley–Hamilton to the generator $A$ on $\mathcal{W}(\psi)$ shows that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ annihilates every orbit $t \mapsto \ell(\exp(tA)v)$, so $\chi_A(D)\psi = 0$ on $[0, \infty)$. Conversely, for $\psi(t) = \sum_i P_i(t) e^{\lambda_i t}$ with distinct rates $\lambda_i$ and $P_i \neq 0$, Hasse-derivative independence shows that $q(D)\psi = 0$ on $[0, \infty)$ if and only if the product annihilator

$$p(X) = \prod_i (X - \lambda_i)^{\deg P_i + 1}$$

divides $q(X)$ (`differentialOperator_sum_expPoly_eq_zero_iff`). Thus $p \mid \chi_A$, giving $\sum_i (\deg P_i + 1) = \deg p \leq \deg \chi_A = \dim_{\mathbb{C}} \mathcal{W}(\psi)$, while Taylor shift expansion of $P_i(u+t)e^{\lambda_i(u+t)}$ gives the matching upper bound $\dim_{\mathbb{C}} \mathcal{W}(\psi) \leq \sum_i (\deg P_i + 1)$.

# 3 Exact Multi-Way Separable Ranks in $d \geq 2$ Dimensions

In dimension $d \geq 3$, there is no single row-span dimension equal to separableRank$(K)$, and scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ no longer preserves rank. Instead, `MultiComplexRank.lean` and `MultiGaussian.lean` establish exact $d$-way upper bounds by root-of-unity identities and matching lower bounds by mapping functional decompositions onto finite-dimensional multilinear forms.

## 3.1 Bridging function spaces and multilinear tensor rank

Suppose $V$ is a $\mathbb{K}$-vector space over a field $\mathbb{K}$, $v : \alpha \to V$ is an evaluation map whose image spans $V$ ($\mathrm{span}_{\mathbb{K}}(\mathrm{range}\, v) = \top$), and $M : V^d \to \mathbb{K}$ is a $d$-multilinear map (`MultilinearMap` $\mathbb{K}$ (Fin $d \to V$) $\mathbb{K}$). Lemma `exists_linear_factors_of_span_eq_top` proves that if the composed kernel $s \in \alpha^d \mapsto M(v(s_0), \ldots, v(s_{d-1}))$ admits a functional decomposition (1) of rank $R$ with arbitrary set-theoretic factors $f_{n,j} : \alpha \to \mathbb{K}$, then there exist *linear* functionals $\ell_{n,j} \in V^*$ such that

$$M(u_0, \ldots, u_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(u_j) \qquad \text{for all } u_0, \ldots, u_{d-1} \in V. \tag{5}$$

Indeed, selecting a $\mathbb{K}$-basis $\{v(s_b)\}_{b \in B}$ of $V$ from range$(v)$ defines unique linear functionals $\ell_{n,j} \in V^*$ satisfying $\ell_{n,j}(v(s_b)) = f_{n,j}(s_b)$; both sides of (5) are $d$-multilinear maps that agree

on every $d$-tuple of basis vectors, so they agree on $V^d$. To verify $\operatorname{span}_{\mathbb{K}}(\operatorname{range} v) = \top$ when $\dim_{\mathbb{K}} V < \infty$, Lemma `span_range_eq_top_of_dual` shows by quotient duality that it suffices to check $L \circ v = 0 \implies L = 0$ for every $L \in V^*$.

### 3.2 Exact $d$-way complex exponential-polynomial rank

Fix $d \geq 2$, a finite index set $\iota$, injective rates $\lambda : \iota \to \mathbb{C}$, and nonzero polynomials $P_i \in \mathbb{C}[X]$ of degree $k_i = \deg P_i$ $(i \in \iota)$.

**Upper bound via shifted root-of-unity interpolation.** For a single polynomial $P \in \mathbb{C}[X]$ of degree $k = \deg P$, put $L = (d-1)k + 1 \geq 1$ and $\zeta = \exp(2\pi i/L)$. Using the $q$-th Hasse derivative $D^{[q]}P$ (so $P(s_0 + t) = \sum_{q=0}^{k} (D^{[q]}P)(s_0)\, t^q$), define the reversed Hasse polynomial on coordinate 0 and the truncated exponential polynomials on coordinates $j \in \{1, \dots, d-1\}$ (`revHassePoly`, `truncExpPoly`):

$$R(s_0; X) = \sum_{q=0}^{k} q!\, (D^{[q]}P)(s_0)\, X^{k-q}, \qquad E_k(s_j; X) = \sum_{q=0}^{k} \frac{s_j^q}{q!} X^q,$$

$$Q(X) = R(s_0; X) \prod_{j=1}^{d-1} E_k(s_j; X).$$

Binomial convolution gives $[X^q] \prod_{j=1}^{d-1} E_k(s_j; X) = (s_1 + \cdots + s_{d-1})^q / q!$ for all $0 \leq q \leq k$, so the coefficient of $X^k$ in $Q(X)$ is precisely $P(s_0 + s_1 + \cdots + s_{d-1})$. Crucially, because $d \geq 2$, we have $k < L$ and $\deg Q \leq k + (d-1)k = k + L - 1 < k + L$. For *any* polynomial $Q$ with $\deg Q < k + L$ and $k < L$, discrete Fourier averaging over only $L$ roots of unity extracts $[X^k]Q(X)$ (`coeff_eq_sum_rootsOfUnity`):

$$[X^k]Q(X) = \frac{1}{L} \sum_{r=0}^{L-1} \zeta^{-rk}\, Q(\zeta^r),$$

because every monomial index $n \in \{0, \dots, k+L-1\}$ satisfies $-L < -k \leq n - k < L$, so $L \mid (n-k)$ holds if and only if $n = k$. Multiplying the $j$-th factor of $Q(\zeta^r)$ by $e^{\lambda s_j}$ (and absorbing $\frac{1}{L}\zeta^{-rk}$ into coordinate 0) expresses $P(\sum_{j=0}^{d-1} s_j) e^{\lambda \sum_{j=0}^{d-1} s_j}$ as a sum of $L = (d-1)\deg P + 1$ coordinatewise products (`isSumOfProducts_expPoly`).

**Lower bound via Alder–Strassen degree peeling.** Let $m_0(i) = \deg P_i + 1$ and form the coefficient evaluation space $V = \prod_{i \in \iota} \mathbb{C}^{m_0(i)}$ with $v(s)_{i,q} = \frac{s^q}{q!} e^{\lambda_i s}$ $(0 \leq q < m_0(i))$. By univariate exponential-polynomial uniqueness (`eq_zero_of_sum_expPoly_eq_zero'`), no nonzero functional in $V^*$ vanishes on range$(v)$, so $\operatorname{span}_{\mathbb{C}}(\operatorname{range} v) = \top$ (`span_expMonomial_eq_top`). Under $v$, the Hankel kernel corresponds to the $d$-multilinear truncated-convolution form (`truncForm`) with cutoffs $c_i \leq m_0(i)$,

$$\mathcal{T}_c(u^{(0)}, \dots, u^{(d-1)}) = \sum_{i \in \iota} \sum_{q_0 + \cdots + q_{d-1} < c_i} w_i(q_0 + \cdots + q_{d-1}) \prod_{j=0}^{d-1} u^{(j)}_{i,q_j},$$

where $w_i(c_i - 1) \neq 0$ whenever $c_i > 0$. Lemma `truncForm_rank_ge` proves

$$\sum_{i:\, c_i > 0} \big((d-1)(c_i - 1) + 1\big) \leq R$$

for every rank-$R$ decomposition (5) by strong induction on $S = \sum_{i \in \iota} c_i$:

- *Case $c_{i_0} = 1$:* Evaluating $\mathcal{T}_c$ at $(e_{i_0,0}, \ldots, e_{i_0,0})$ gives $w_{i_0}(0) \neq 0$, selecting an index $n_0$ with $\ell_{n_0,0}(e_{i_0,0}) \neq 0$. Let $\pi_{\neq i_0} : V \to V$ zero out block $i_0$. Substituting $\varphi_0(x) = \pi_{\neq i_0}(x) - \frac{\ell_{n_0,0}(\pi_{\neq i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 and $\pi_{\neq i_0}$ on modes $j \geq 1$ annihilates $\ell_{n_0,0} \circ \varphi_0$ while reducing $c_{i_0}$ from 1 to 0 (since $d \geq 2$ zeroes block $i_0$ on mode 1), leaving at most $R - 1$ terms.
- *Case $c_{i_0} \geq 2$:* Set $s_{\text{top}} = e_{i_0,c_{i_0}-1}$ and $s_0 = e_{i_0,0}$ ($s_{\text{top}} \neq s_0$). Because $(c_{i_0} - 1) + (c_{i_0} - 1) \geq c_{i_0}$, any rank-1 perturbation of the form $\psi(x) = x - \gamma(x) s_{\text{top}}$ leaves $\mathcal{T}_c$ unchanged whenever another coordinate equals $s_{\text{top}}$. Stepping through modes $j = 0, \ldots, d-2$ sequentially evaluates $\mathcal{T}_c$ with $u^{(j)} = s_{\text{top}}$ and $u^{(k)} = \psi_k(s_0)$ ($k \neq j$) to find $d - 1$ distinct indices $D \subseteq \operatorname{Fin} R$ ($|D| = d - 1$) killed on modes $0, \ldots, d-2$ by rank-1 perturbations $\psi_j$ along $s_{\text{top}}$. On modes $0, \ldots, d-2$ we then truncate block $i_0$ to indices $q < c_{i_0} - 1$ before applying $\psi_j$, while on mode $d - 1$ we shift block $i_0$'s degree indices up by $+1$ ($e_{i_0,q} \mapsto e_{i_0,q+1}$ for $q < c_{i_0} - 1$): this transforms $\mathcal{T}_c$ into the reduced form $\mathcal{T}_{c'}$ ($c' = c[i_0 \mapsto c_{i_0} - 1]$) with shifted weights $w'_{i_0}(m) = w_{i_0}(m+1)$—preserving the nonzero top weight $w'_{i_0}(c'_{i_0} - 1) = w_{i_0}(c_{i_0} - 1) \neq 0$—while dropping all $d - 1$ terms in $D$.

Combining this lower bound at $c = m_0$ with the upper bound and Section 2 gives:

**Theorem 3.1** ($d$-D Complex Radial Rank, `separableRank_radial_complex_eq_iff`)**.** *Let $d \geq 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0, \infty)$. For every $N \in \mathbb{N}$,*

$$\operatorname{separableRank}\left( x \in \mathbb{R}^d \mapsto \psi\left( \sum_{j=0}^{d-1} x_j^2 \right) \right) = N$$

*if and only if there exist a finite set $S \subset \mathbb{C}$ and polynomials $(P_\mu)_{\mu \in S}$ with $P_\mu \neq 0$ such that $\psi(s) = \sum_{\mu \in S} P_\mu(s) e^{\mu s}$ for all $s \geq 0$ and $\sum_{\mu \in S} \big( (d-1) \deg P_\mu + 1 \big) = N$.*

### 3.3 Exact $d$-way real multi-Gaussian rank and root-of-unity real decomposition

Over $\mathbb{R}$, expanding each factor of $\operatorname{Re}(\prod_{j=0}^{d-1} z_j)$ into $\operatorname{Re} z_j + i \operatorname{Im} z_j$ produces $2^{d-1}$ real products where an even number of imaginary parts are chosen. Remarkably, $d$ real products always suffice:

**Proposition 3.2** (Alternating $2d$-th root-of-unity identity, `re_prod_eq_sum_re`)**.** *For every integer $d \geq 1$ and all $z_0, \ldots, z_{d-1} \in \mathbb{C}$,*

$$\operatorname{Re}\left( \prod_{j=0}^{d-1} z_j \right) = \frac{2^{d-1}}{d} \sum_{m=0}^{d-1} (-1)^m \prod_{j=0}^{d-1} \operatorname{Re}\left( z_j e^{-i\pi m/d} \right). \tag{6}$$

*Proof.* Write $\operatorname{Re}(z_j e^{-i\pi m/d}) = \frac{1}{2}\left( \overline{z_j} e^{i\pi m/d} + z_j e^{-i\pi m/d} \right)$ and expand the product over subsets $S \subseteq \{0, \ldots, d-1\}$:

$$\prod_{j=0}^{d-1} \operatorname{Re}\left( z_j e^{-i\pi m/d} \right) = 2^{-d} \sum_{S \subseteq \{0,\ldots,d-1\}} \left( \prod_{j \in S} \overline{z_j} \right) \left( \prod_{j \notin S} z_j \right) e^{i\pi m(2|S|-d)/d}.$$

Since $(-1)^m = e^{i\pi m}$, multiplying by $(-1)^m$ turns the $m$-dependent factor for subset $S$ into $w_S^m$, where $w_S = e^{2\pi i |S|/d}$ satisfies $w_S^d = 1$. When $S = \emptyset$ or $S = \{0, \ldots, d-1\}$, $w_S = 1$ and $\sum_{m=0}^{d-1} w_S^m = d$. When $0 < |S| < d$, $w_S \neq 1$, so $(w_S - 1) \sum_{m=0}^{d-1} w_S^m = w_S^d - 1 = 0$ kills the sum. Only $S = \emptyset$ and $S = \{0, \ldots, d-1\}$ survive, giving

$$\frac{2^{d-1}}{d} \cdot 2^{-d} \cdot d \left( \prod_{j=0}^{d-1} z_j + \prod_{j=0}^{d-1} \overline{z_j} \right) = \frac{1}{2} \left( \prod_{j=0}^{d-1} z_j + \overline{\prod_{j=0}^{d-1} z_j} \right) = \operatorname{Re}\left( \prod_{j=0}^{d-1} z_j \right). \qquad \square$$

Absorbing $\frac{2^{d-1}}{d}(-1)^m$ into the $j=0$ factor shows that taking the real part of any rank-$N$ complex $d$-way kernel yields a real $d$-way kernel of rank at most $Nd$ (`IsSumOfProducts.re`). Together with Theorem 3.1 and the slice bound (4), this completes the $d$-dimensional qualitative classification:

**Theorem 3.3** ($d$-D Finite-Rank Classification, `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0,\infty)$. Then*

$$\operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) < \infty$$
$$\iff \exists\, \mathrm{cs},\ \operatorname{IsRepresentation}(\mathrm{cs}) \wedge \forall s \geq 0,\ \varphi(s) = \operatorname{profile}(\mathrm{cs}, s).$$

For a single nonreal Gaussian component $\operatorname{Re}(\beta e^{\lambda \sum_{j=0}^{d-1} x_j^2})$ ($\beta \neq 0$, $\operatorname{Im}\lambda > 0$), setting $z_0 = \beta e^{\lambda x_0^2}$ and $z_j = e^{\lambda x_j^2}$ ($j \geq 1$) in Proposition 3.2 yields a real $d$-way decomposition of rank $d$ (`isSumOfProducts_radial_gaussianComponent`). Thus a Gaussian representation with $r$ real rates and $c$ nonreal rates has $d$-way real separable rank at most $r + dc$.

To prove the matching lower bound $R \geq r + dc$ when $d \geq 2$ (`MultiGaussian.lean`), consider the real vector space $G = \mathbb{R}^{\iota_{\mathbb{R}}} \times \mathbb{C}^{\iota_{\mathbb{C}}}$, where $\iota_{\mathbb{R}}$ indexes the real rates $\alpha_i e^{\mu_i s}$ ($\alpha_i \in \mathbb{R} \setminus \{0\}$) and $\iota_{\mathbb{C}}$ indexes the upper-half-plane rates $\operatorname{Re}(\beta_k e^{\lambda_k s})$ ($\beta_k \in \mathbb{C} \setminus \{0\}$, $\operatorname{Im}\lambda_k > 0$). By conjugate-rate uniqueness, the evaluation curve $v(s) = \big((e^{\mu_i s})_i, (e^{\lambda_k s})_k\big)$ ($s \in \mathbb{R}_{\geq 0}$) spans $G$ over $\mathbb{R}$ (`span_gaussianEval_eq_top`), transferring any rank-$R$ decomposition of the Hankel kernel to a rank-$R$ $\mathbb{R}$-multilinear decomposition (5) of

$$\mathcal{G}(u^{(0)}, \ldots, u^{(d-1)}) = \sum_{i \in S_{\mathbb{R}}} \alpha_i \prod_{j=0}^{d-1} u_{\mathbb{R},i}^{(j)} + \sum_{k \in S_{\mathbb{C}}} \operatorname{Re}\left(\beta_k \prod_{j=0}^{d-1} u_{\mathbb{C},k}^{(j)}\right).$$

Two geometric facts over $\mathbb{R}$ drive the lower-bound induction (`single_complex_rank_ge`, `gaussianForm_rank_ge`):

1. **Single complex rate (`single_complex_rank_ge`):** Every $\mathbb{R}$-linear functional $L : \mathbb{C} \to \mathbb{R}$ has a nonzero kernel vector $w \in \ker L \setminus \{0\}$ (`exists_ker_ne_zero`: take $w = 1$ if $L(1) = L(i) = 0$, else $w = L(i) - i\,L(1)$). We prove $\operatorname{Re}(\beta \prod_{j=0}^{d-1} z_j) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(z_j) \implies d \leq R$ ($\beta \neq 0$) by induction on $d \geq 2$. For $d = 2$, $R = 0$ fails at $(1, \bar{\beta})$, and if $R = 1$, picking $w \in \ker(\ell_{0,0}) \setminus \{0\}$ and evaluating at $(w, \overline{\beta w})$ yields $|\beta w|^2 > 0$ on the left and 0 on the right. For $d \mapsto d+1$ with $R \geq 1$, picking $w \in \ker(\ell_{R-1,d}) \setminus \{0\}$ on the last coordinate kills the $(R-1)$-st product while replacing $\beta$ by $\beta w \neq 0$ on the first $d$ coordinates, so the induction hypothesis gives $d \leq R - 1$.

2. **Peeling real rates (`gaussianForm_real_rank_ge`):** When $S_{\mathbb{R}}$ is nonempty, picking $i_0 \in S_{\mathbb{R}}$ and evaluating at $e_{\mathbb{R},i_0}$ finds an index $n_0$ nonzero at $e_{\mathbb{R},i_0}$; subtracting a rank-1 multiple of $e_{\mathbb{R},i_0}$ from the projection $\pi_{\neq i_0}$ on mode 0 (and applying $\pi_{\neq i_0}$ on modes $j \geq 1$) kills term $n_0$ and removes $i_0$ from $S_{\mathbb{R}}$, reducing $R \mapsto R - 1$.

3. **Peeling complex rates (`gaussianForm_rank_ge`):** Once $S_{\mathbb{R}} = \emptyset$, induct on $|S_{\mathbb{C}}|$. Given $k_0 \in S_{\mathbb{C}}$, let $\operatorname{emb}_{k_0} : \mathbb{C} \to G$ embed into slot $k_0$ and $\pi_{\neq k_0} : G \to G$ zero out slot $k_0$. On mode 0, evaluating at $1_{k_0} = \operatorname{emb}_{k_0}(1)$ selects $n_1$ with $\ell_{n_1,0}(1_{k_0}) \neq 0$, and evaluating at $w_1 \in \ker(\ell_{n_1,0} \circ \operatorname{emb}_{k_0}) \setminus \{0\}$ selects a second index $n_2 \neq n_1$ with $\ell_{n_2,0}(w_1) \neq 0$; subtracting sequential rank-1 projections along $1_{k_0}$ and $w_1$ constructs $\psi_0 = \pi_{\neq k_0} - \operatorname{emb}_{k_0} \circ c_0$ annihilating *both* $\ell_{n_1,0}$ and $\ell_{n_2,0}$. Next, for each mode $j = 1, \ldots, d-2$, if every remaining index $n \notin D$ vanished on $\operatorname{emb}_{k_0}(\mathbb{C})$ in mode $j$, then restricting all $d$ coordinates to $\operatorname{emb}_{k_0}(\mathbb{C})$ would represent $\operatorname{Re}(\beta_{k_0} \prod_{q=0}^{d-1} z_q)$ with $|D| = j + 1 < d$ terms, contradicting Step 1. Hence each mode

$1, \dots, d-2$ yields one more distinct index $n_r \notin D$ killed by a map $\psi_j = \pi_{\neq k_0} - \mathrm{emb}_{k_0} \circ c_j$, giving $|D| = d$ killed terms across modes $0, \dots, d-2$. Finally, setting $\psi_{d-1} = \pi_{\neq k_0}$ on the last mode $d-1 \geq 1$ ensures that for every $x$, $(\psi_0(x_0), \dots, \psi_{d-1}(x_{d-1}))$ has zero $k_0$-component on mode $d-1$, reducing $S_{\mathbb{C}} \mapsto S_{\mathbb{C}} \setminus \{k_0\}$ with at most $R - d$ surviving terms.

**Theorem 3.4** ($d$-D Real Multi-Gaussian Rank, `separableRank_radial_gaussian_profile`)**.** *Let $d \geq 2$ and let* cs *be a Gaussian representation (`IsGaussianRepresentation`) with $r$ real rates and $c$ nonreal upper-half-plane rates. Then*

$$\mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \mathrm{profile}\left(\mathrm{cs}, \sum_{j=0}^{d-1} x_j^2\right)\right) = \mathrm{gaussianTotalCost}(d, \mathrm{cs}) = r + dc.$$

**Corollary 3.5** (Failure of real-to-complex rank invariance for $d \geq 3$)**.** *For every $d \geq 3$, the real radial kernel $K(x) = \mathrm{Re}\big(\exp(i \sum_{j=0}^{d-1} x_j^2)\big)$ on $\mathbb{R}^d$ satisfies* $\mathrm{separableRank}_{\mathbb{R}}(K) = d$ *by Theorem 3.4 ($r = 0, c = 1$), whereas its complexification $K_{\mathbb{C}}(x) = \frac{1}{2}e^{i\|x\|^2} + \frac{1}{2}e^{-i\|x\|^2}$ satisfies* $\mathrm{separableRank}_{\mathbb{C}}(K_{\mathbb{C}}) = 2$ *by Theorem 3.1.*

In two dimensions ($d = 2$), real-to-complex scalar extension preserves rank unconditionally (`separableRank_ofReal`), yielding exact real ranks for arbitrary polynomial degrees alongside the minimal differential-annihilator theorem:

**Theorem 3.6** (2-D Exact Real Rank, `separableRank_radial_eq_iff`)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $N \in \mathbb{N}$,*

$$\begin{aligned}&\mathrm{separableRank}\big(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)\big) = N \\ &\quad \iff \exists\, \mathrm{cs},\ \mathrm{IsRepresentation}(\mathrm{cs}) \wedge (\forall s \geq 0,\ \varphi(s) = \mathrm{profile}(\mathrm{cs}, s)) \wedge \mathrm{totalCost}(\mathrm{cs}) = N.\end{aligned}$$

**Theorem 3.7** (Minimal Differential Annihilator)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with finite-dimensional translate span and two-dimensional radial rank* $\mathrm{separableRank}(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = N < \infty$ *(`exists_minimal_differential_annihilator`). There exists a monic polynomial $p \in \mathbb{C}[X]$ of degree $N$ such that for every $q \in \mathbb{C}[X]$, the half-line differential equation $q(D)\varphi_{\mathbb{C}} = 0$ holds on $[0, \infty)$ (where $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$) if and only if $p \mid q$.*

**Remark 3.8** (Open problem for $d \geq 3$ and $\deg P \geq 1$ over $\mathbb{R}$)**.** While Theorem 3.3 characterizes finiteness for all polynomial degrees and Theorem 3.1 determines the exact complex $d$-way rank $\sum_\mu((d-1)\deg P_\mu + 1)$ for all degrees, determining the exact *real* $d$-way rank when $d \geq 3$ and $\deg P_i \geq 1$ for nonreal rates reduces to the real symmetric/Hankel tensor rank of $\mathrm{Re}(P(\sum_j s_j)e^{\lambda \sum_j s_j})$, where real Waring ranks depend delicately on root configurations and exceed complex ranks.

# 4 Standard vs. Nonstandard Proofs of the Semigroup Step

Every proof of necessity above rests on the continuous one-parameter semigroup theorem: if $\mathcal{A}$ is a Banach algebra over $\mathbb{R}$ and $M : \mathbb{R} \to \mathcal{A}$ satisfies $M(0) = 1$, $M(s+t) = M(s)M(t)$ for all $s, t \geq 0$, and $M$ is continuous on $[0, \infty)$, then there exists $A \in \mathcal{A}$ such that $M(t) = \exp(tA)$ for all $t \geq 0$.

## 4.1 Integral smoothing vs. Euler's infinitesimal step with homogeneous scale

**Integral smoothing (`Semigroup.lean`).** Since $M(0) = 1$ and $M$ is continuous at 0, there exists $\delta > 0$ such that $I_\delta = \int_0^\delta M(u)\,du$ satisfies $\|I_\delta - \delta \cdot 1\| < \delta$, making $I_\delta$ invertible in $\mathcal{A}$ by Neumann series. For $t \geq 0$, changing variables in $M(t)I_\delta = \int_t^{t+\delta} M(u)\,du$ shows by the fundamental theorem of calculus that $M$ is right-differentiable on $[0, \infty)$ with $M'(t) = M(t)A$, where $A = (M(\delta) - 1)I_\delta^{-1}$ commutes with $M(t)$. Differentiating $t \mapsto M(t)\exp(-tA)$ gives 0, so $M(t) = \exp(tA)$ in *any* Banach algebra.

**Finite differences (`GeneratorBound.lean`, `FiniteDifferenceSemigroup.lean`).** Without integration, if $\|M(u) - 1\| \le \frac{1}{2}$ on $[0, \delta]$, geometric-series telescoping bounds the difference quotients $\|(M(\delta/n) - 1)/(\delta/n)\|$. In finite dimensions, compactness extracts a convergent subsequence $(M(h_k) - 1)/h_k \to A$; along $h_k$, the one-sided difference quotients of $M(t)\exp(-tA)$ tend to 0, and a one-sided mean-value lemma shows $M(t)\exp(-tA) = 1$.

**Nonstandard analysis with homogeneous common scale.** Euler's heuristic sets $A = \mathrm{st}((M(\varepsilon) - 1)/\varepsilon)$ for a positive infinitesimal $\varepsilon > 0$ and iterates $M(\varepsilon)^{\lfloor t/\varepsilon \rfloor} \approx (1 + \varepsilon A)^{\lfloor t/\varepsilon \rfloor} \approx \exp(tA)$. Directly applying st to $(M(\varepsilon) - 1)/\varepsilon$, however, requires proving beforehand that this quotient is finite (the 124-line geometric-sum bound above).

All three nonstandard formalizations in this project eliminate that *a priori* bound by passing to *projective / homogeneous coordinates*: put $x = M(\varepsilon) - 1$ (which is infinitesimal by continuity at 0) and define the common infinitesimal scale

$$\eta = \varepsilon + \|x\| > 0. \tag{7}$$

Because $\frac{\varepsilon}{\eta} + \left\|\frac{x}{\eta}\right\| = 1$ with $\frac{\varepsilon}{\eta} \in [0, 1]$ and $\left\|\frac{x}{\eta}\right\| \le 1$, both rescaled quantities are bounded by 1 *without any analytical estimate*. In a finite-dimensional algebra $\mathcal{A}$, Heine–Borel compactness gives a standard vector $B = \mathrm{st}(x/\eta)$ with $\|B\| \le 1$ and $\varepsilon/\eta \approx a := 1 - \|B\| \in [0, 1]$. Now compare $N = \lfloor s/\eta \rfloor$ steps of $M(\varepsilon)$ (arriving at time $N\varepsilon \approx sa$) against $N$ steps of $\exp(\eta B)$ (arriving at time $N\eta \approx s$) for an arbitrary standard $s \ge 0$:

$$\frac{\|M(\varepsilon) - \exp(\eta B)\|}{\eta} \le \left\|\frac{x}{\eta} - B\right\| + \left\|B - \frac{\exp(\eta B) - 1}{\eta}\right\| \approx 0.$$

Summing $N$ single-step errors of size $\eta \cdot o(1)$ with $N\eta \le s$ yields $M(sa) = \exp(sB)$ for all standard $s \ge 0$. Could the time coordinate vanish ($a = 0$)? If $a = 0$, then $\exp(sB) = M(0) = 1$ for every $s \ge 0$; differentiating at $s = 0$ gives $B = 0$, contradicting $a + \|B\| = 1$! Hence $a > 0$, and setting $A = B/a$ with $s = t/a$ gives $M(t) = \exp(tA)$.

### 4.2 Three nonstandard formalizations and quantitative comparison

We formalized the homogeneous-scale nonstandard proof in three settings:

1. **Isabelle/HOL:** `Isabelle/Semigroup_NSA.thy` uses `HOL-Nonstandard_Analysis`, which provides hyperreals, hypernaturals, star extensions `*f*`, and the `transfer` proof method. Single-step error accumulation reduces to a purely algebraic noncommutative power estimate (`power_difference_bound`): if $\|u\|, \|v\| \le e^h$ with $h \ge 0$, then $\|u^n - v^n\| \le ne^{nh}\|u - v\|$, proved by induction from $u^{n+1} - v^{n+1} = u(u^n - v^n) + (u - v)v^n$. Since $\|M(\varepsilon)\| \le 1 + \|x\| \le 1 + \eta \le e^\eta$ and $\|\exp(\eta B)\| \le e^{\eta\|B\|} \le e^\eta$, `transfer` applies this ordinary inequality directly to a hypernatural step count $N$. (This uses $\|1\| = 1$ from Isabelle's `real_normed_algebra_1`; in Lean, $\mathcal{L}(\mathcal{W}(\psi))$ has $\|1\| = 0$ when $\psi = 0$, so the Lean theorems avoid assuming $\|1\| = 1$ and telescope steps using compact-interval bounds via `norm_replace_steps_le` instead.)

2. **Lean 4 with ultrafilter sequences:** `Experiments/NonstandardSemigroup.lean` expresses "infinitely close" directly as convergence along hyperfilter $\mathbb{N}$ (since Mathlib defines Hyperreal = Germ(hyperfilter $\mathbb{N}$) $\mathbb{R}$ without a transfer tactic), obtaining $B$ in one line via `IsCompact.ultrafilter_le_nhds`.

3. **Lean 4 in nonstandard vocabulary:** In `Experiments/`, `Ultrapower.lean` and `UltrapowerSemigroup.lean` build a reusable 195-line ultrapower interface (`Hyp`, `std`, `ext`, $\approx_{\mathrm{h}}$, one-variable transfer lemmas) and write Euler's proof in classical nonstandard notation.

Table 1: Verified source sizes for the five complete proofs of the continuous one-parameter semigroup step $M(t) = \exp(tA)$. Every project helper lemma in each proof's module dependencies is counted in full (excluding standard Mathlib and Isabelle/HOL library imports).

| Proof method | Prover / files | Phys. | Nonbl. | Scope |
|---|---|---|---|---|
| Integral smoothing (main) | Lean 4 (`Semigroup.lean`) | 96 | 87 | Any Banach algebra |
| Finite differences | Lean 4 (`GeneratorBound`, `FD`) | 256 | 233 | Finite-dim. |
| NSA (ultrafilter seq.) | Lean 4 (`NonstandardSemigroup`) | 206 | 190 | Finite-dim. |
| NSA (ultrapower API) | Lean 4 (`Ultrapower*` + shared) | 399 | 344 | Finite-dim. |
| NSA (star + `transfer`) | Isabelle (`Semigroup_NSA.thy`) | 275 | 253 | Finite-dim. ($\|1\| = 1$) |

Table 1 summarizes the exact line counts. Although the homogeneous-scale nonstandard proof uses no integration and only elementary algebra, *integral smoothing remains both the shortest (96 lines) and the only infinite-dimensional proof*, because Mathlib provides Bochner integration and the fundamental theorem of calculus as off-the-shelf library theorems. Even discounting the 195-line interface `Ultrapower.lean` as reusable library code, the nonstandard-language Lean proof ($169 + 35 = 204$ lines) merely matches the direct ultrafilter proof (206 lines): without automated transfer, nonstandard vocabulary in Lean just wraps filter limits. In Isabelle/HOL, `transfer` genuinely compresses hypernatural induction, but 59 lines of missing library lemmas (Heine–Borel vector standard parts and half-line continuity via $M(\max(t, 0))$) plus stating transferred formulas at both standard and nonstandard types keep the total at 275 lines.

# 5 Applications to Isotropic and Steerable Filtering and 3-D Propagation

## 5.1 Harmonic "beach-ball" differential operators and steerable kernels

Specializing Proposition 3.2 to equal symbolic factors $z_0 = \cdots = z_{d-1} = \partial_x + i\partial_y$ (writing $m$ for the harmonic order and $D_\alpha = \cos\alpha\,\partial_x + \sin\alpha\,\partial_y$ for the directional derivative along angle $\alpha$) gives, for *every* integer $m \geq 1$ (both odd and even),

$$\operatorname{Re}\big((\partial_x + i\partial_y)^m\big) = \frac{2^{m-1}}{m} \sum_{k=0}^{m-1} (-1)^k D^m_{\pi k/m}. \tag{8}$$

On a regular $2m$-point ring $\theta_n = \pi n/m$ ($n = 0, \ldots, 2m-1$) of radius $h \to 0$, antipodal points satisfy $D_{\theta_{k+m}} = -D_{\theta_k}$ while Fourier orthogonality of $(-1)^k = e^{i\pi k}$ annihilates all Taylor orders below $m$. For $f \in C^m$ near $(x, y)$, (8) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order ($m = 2N + 1$) hexagonal/decagonal "beach-ball" operators of Niemitalo [7] and the even-order operators of Perona [10] in a single formula:

$$\lim_{h\to 0} \frac{1}{h^m} \sum_{n=0}^{2m-1} (-1)^n f\Big(x + h\cos\frac{\pi n}{m},\, y + h\sin\frac{\pi n}{m}\Big) = \frac{1}{(m-1)!\,2^{m-2}} \operatorname{Re}\big((\partial_x + i\partial_y)^m\big) f(x, y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\operatorname{Im}((\partial_x + i\partial_y)^m)$. Recognizing $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial^m_{\bar z}$ to *any* smooth radial kernel $\varphi(x^2 + y^2) = \varphi(z\bar z)$ produces no lower-order radial derivatives: because $\partial_{\bar z}(z) = 0$, induction gives

$$\left(\frac{\partial_x + i\partial_y}{2}\right)^m \varphi(x^2 + y^2) = (x + iy)^m\, \varphi^{(m)}(x^2 + y^2) = r^m e^{im\theta}\, \varphi^{(m)}(r^2). \tag{9}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with $r$ real and $c$ nonreal upper-half-plane rates (all $\lambda_i \neq 0$, approximating, e.g., an ideal circular low-pass filter [8]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by $\lambda_i^m$) of continuous 2-way rank $R = r+2c$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [6]—absorbs the monomials of $\mathrm{Re}((x+iy)^m)$ and $\mathrm{Im}((x+iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2 \rfloor + 1\big)(r+2c) \quad \text{for } \cos(m\theta), \qquad \lceil m/2 \rceil (r+2c) \quad \text{for } \sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms (grid aliasing at special rates such as $\mathrm{Im}\,\lambda_i \in \pi\mathbb{Z}$ can collapse the sampled discrete rank further).

**Continuous-domain pass-count bounds for simultaneous steerable pairs.** In applications such as orientation estimation [6], one evaluates both continuous-domain convolutions $(C_m * I, S_m * I)$ of a harmonic pair simultaneously. Can sharing intermediate 1-D continuous convolutions reduce the pass count below separate evaluation $2(\lfloor m/2 \rfloor + 1)R + 2\lceil m/2 \rceil R = 2(m+1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m, S_m) = (A_c + B_c, A_s + B_s)$ into $P_y$ shared vertical passes followed by $\mathrm{rank}(A_c) + \mathrm{rank}(A_s)$ horizontal passes plus $P_x$ shared horizontal passes followed by $\mathrm{rank}(B_c) + \mathrm{rank}(B_s)$ vertical passes, then

$$\dim \mathrm{rowSpan}_x(C_m, S_m) \leq \mathrm{rank}(A_c) + \mathrm{rank}(A_s) + P_x,$$
$$\dim \mathrm{colSpan}_y(C_m, S_m) \leq P_y + \mathrm{rank}(B_c) + \mathrm{rank}(B_s).$$

Because $C_m$ and $S_m$ on $\mathbb{R}^2$ span all powers $\{x^0, \ldots, x^m\} g_n(x)$ in $x$ and $\{y^0, \ldots, y^m\} g_n(y)$ in $y$ (for $m = 1$, $h_x$ is odd-in-$x$/even-in-$y$ while $h_y$ is even-in-$x$/odd-in-$y$), adding the two inequalities proves that *every* two-stage continuous 1-D schedule requires at least $(m+1)R + (m+1)R = 2(m+1)R$ passes ($4R$ passes for $(h_x, h_y)$). Savings instead arise in three ways: (i) definite even/odd parity of every 1-D factor $u^k g_n(u)$ halves the multiplications per sampled pass via $I[y+\ell] \pm I[y-\ell]$ folding while fusing input loads; (ii) across *multiple* harmonic orders $m = 0, \ldots, M$, vertical passes $\{I *_y (y^k g_n)\}_{k=0}^{M}$ are shared linearly ($5R$ passes instead of $6R$ for low-pass plus gradient; $10R$ instead of $12R$ for $m = 1$ plus $m = 3$); and (iii) for the $d$-dimensional gradient $(h_{x_0}, \ldots, h_{x_{d-1}})$ of a multi-Gaussian radial kernel of $d$-way rank $R_d = r + dc$ ($d \geq 3$), recursively splitting $\{0, \ldots, d-1\}$ into halves of sizes $\lfloor d/2 \rfloor$ and $\lceil d/2 \rceil$ and pre-convolving each half with the undifferentiated factors reduces $d^2 R_d$ passes to $T(d)R_d$ with $T(1) = 1$ and $T(d) = T(\lfloor d/2 \rfloor) + T(\lceil d/2 \rceil) + d \leq d(\lceil \log_2 d \rceil + 1)$ ($8R_3$ vs. $9R_3$ in $d = 3$; $12R_4$ vs. $16R_4$ in $d = 4$).

### 5.2 Three-dimensional filtering and band-limited wave propagation

In three dimensions ($d = 3$), direct spatial convolution with an $L \times L \times L$ isotropic kernel—such as a cryo-EM/tomography bandpass filter or a coarse-grid dispersion-free time-step propagator for the band-limited constant-speed acoustic wave equation, whose spatial-frequency symbol depends radially on $\|\xi\|^2$—scales as $L^3$ per voxel. Approximating the radial profile by $r$ real and $c$ complex-conjugate Gaussians yields exact real 3-way separable rank $r+3c$ by Theorem 3.4: Proposition 3.2 realizes each complex Gaussian in 3 real 3-way terms (9 real 1-D passes on real-valued volume buffers, vs. $2^{3-1} = 4$ terms / 12 passes from naive real-and-imaginary expansion), and certifies that no real 3-way factorization uses fewer than $r + 3c$ terms.

## 6 Formalization Architecture and Verification Audit

Table 2 breaks down the `KernelRankLean` library (3,822 physical lines, 3,537 nonblank lines; the 2-D classification and differential-annihilator suite prior to the $d$-dimensional tensor-rank

Table 2: Structure and source line counts of the `KernelRankLean` library (commit `6aabb4d`).

| Module(s) | Mathematical content | Phys. | Nonbl. |
|---|---|---|---|
| `Defs`, `KernelRankLean` | Multi-way rank, surjections, slices, costs | 183 | 156 |
| `RowSpan`, `Complexification` | 2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance | 221 | 189 |
| `Translations` | $d$-D radial-to-Hankel reduction; `translateSpan` | 92 | 80 |
| `ExpPoly`, `RealExpPoly` | Exponential-polynomial uniqueness | 198 | 170 |
| `Semigroup`, `TranslationSemigroup` | Banach semigroups; shift semigroup | 262 | 219 |
| `Necessity`, `Representation` | Generalized eigenvectors; conjugate pairing | 222 | 198 |
| `Differential`, `ComplexRank` | Half-line operators $q(D)$; 2-D complex rank | 318 | 284 |
| `Sufficiency`, `Annihilator` | 2-D real sufficiency; minimal annihilator | 196 | 181 |
| `MultiComplexRank` | Exact $d$-way complex rank $\sum_i((d-1)\deg P_i + 1)$ | 1,008 | 972 |
| `MultiGaussian` | $2d$-th root identity; exact real rank $r + dc$ | 978 | 955 |
| `Main` | Unified $d$-D and 2-D main theorems | 144 | 133 |
| **Total (`KernelRankLean`)** | | **3,822** | **3,537** |

modules comprised 1,587 physical lines at commit `cae13d6`, and 1,340 physical lines at commit `240dea0` prior to differential annihilators). Three formalization choices proved decisive in Lean 4:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a general dependent family $\prod_{i\in\iota} X_i$ unifies 2-D and $d$-D statements, whereas specializing `rowSpan` in `RowSpan.lean` to homogeneous $\mathrm{Fin}\,2 \to \alpha$ (with a pattern-matched constructor `fin2Vec`) avoids expensive higher-order unification timeouts when rewriting on $\mathrm{Fin}\,2 \to \mathbb{R}_{\geq 0}$.
2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto the finite-dimensional subspace $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting `LinearMap.continuous_of_finiteDimensional` supply automatic continuity.
3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated both coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 3.7 directly.

**Verification audit.** A single script (`scripts/check`, also wired as the repository's pre-commit hook) builds both `KernelRankLean` and `Experiments` via `lake build` (which includes every `.lean` module in the repository), checks the compiler log to ensure no declaration uses `sorry`, runs `#print axioms` on 14 top-level theorems to confirm dependence solely on Lean's three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), builds the Isabelle2026 session `KernelRank_NSA`, and audits `Semigroup_NSA.semigroup_eq_exp` via `Thm_Deps.all_oracles` (catching `skip_proof`/`sorry` even under `quick_and_dirty`) while rejecting any `axiomatization` or `axioms` declarations.

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This work builds directly on problems, mathematical formulations, and software developed by Olli Niemitalo, including `kernel-optimizer` [9] for complex-Gaussian approximations of 2-D circularly symmetric blur kernels and his investigations of isotropic low-pass filters, steerable

gradient orientation estimation, and planar harmonic differential operators [6, 7, 8]. Olli Niemitalo initiated and guided the `kernel-rank-lean` repository, proposed comparing standard and Isabelle/HOL nonstandard proofs of the semigroup step, and suggested extending the formalization to $d$ dimensions.

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